

Lab 28 — Network Fault Diagnosis — Command Line to Packet Level

Course: CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)

Topic 05: Hardware and Network Troubleshooting — Core 1, 28% of Core 1

Exam objective: Diagnose network faults by working the layers in order, from physical connectivity through addressing and DNS to application behaviour (Core 1 objective 5.7).

Goal

Network faults are diagnosed layer by layer, because testing at the wrong layer wastes time and misleads. You work a fixed bottom-up sequence with a specific command at each layer, then use packet analysis to resolve what the commands alone cannot.

What you'll produce

A layered network diagnostic sequence with a command per layer, and a symptom-to-cause map for eight network faults.

Tools and equipment

Killercoda Ubuntu Playground, PCAP Analyzer, IP Calculator, ping, traceroute, dig, ss, tcpdump

Browser tools used in this lab

- **IP Calculator** — <https://alfredang.github.io/ipcalculator/>
- **PCAP Analyzer** — <https://alfredang.github.io/pcapanalyzer/>
- **Killercoda Ubuntu Playground** — <https://killercoda.com/playgrounds/scenario/ubuntu>

IP Subnet Calculator

A modern, sipcalc-inspired calculator for IPv4 and IPv6 with batch input, recursive subnet splitting, and DNS lookups.

IPv4 IPv6 Batch

IPv4 address / netmask

192.168.1.10

/ 24

Calculate

CIDR (24), dotted netmask (255.255.255.0), hex mask (0xffffffff), or paste 192.168.1.10/24 directly.

Network class

Any

Subnet mask

/24 255.255.255.0 (254 hosts)

Quick:

10.0.0.1/8

172.16.5.4/12

192.168.1.10/24

192.168.10.50/26

203.0.113.5/30

BIT VISUALIZER · CLICK ANY BIT TO TOGGLE PREFIX · DRAG SLIDER

Copy share link

192.168.1.10/24

1 1 0 0 0 0 0 0 . 1 0 1 0 1 0 0 0 . 0 0 0 0 0 0 0 1 . 0 0 0 0 1 0 1 0

/0

/32

NETWORK 192.168.1.0

BROADCAST 192.168.1.255

MASK 255.255.255.0

HOSTS 254

ADDRESS (MULTIPLE FORMATS)

Decimal 192.168.1.10

Hex (full) 0xc0a8010a

Hex dotted C0.A8.01.0A

Integer 3232235786

CIDR /24

Class C (default /24)

Type PRIVATE (RFC1918)

NETMASK FORMATS

Dotted 255.255.255.0

Hex 0xffffffff

CIDR Copied

Wildcard 0.0.0.255

Wildcard hex 0x000000ff

NETWORK

Network 192.168.1.0

Network hex 0xc0a80100

Broadcast 192.168.1.255

Broadcast hex 0xc0a801ff

Range 192.168.1.0 - 192.168.1.255

HOST RANGE

First host 192.168.1.1

Last host 192.168.1.254

Total IPs 256

Usable 254 usable

Reverse DNS 0.1.168.192.in-addr.arpa

BITMAPS

IP 11000000.10101000.00000001.00001010

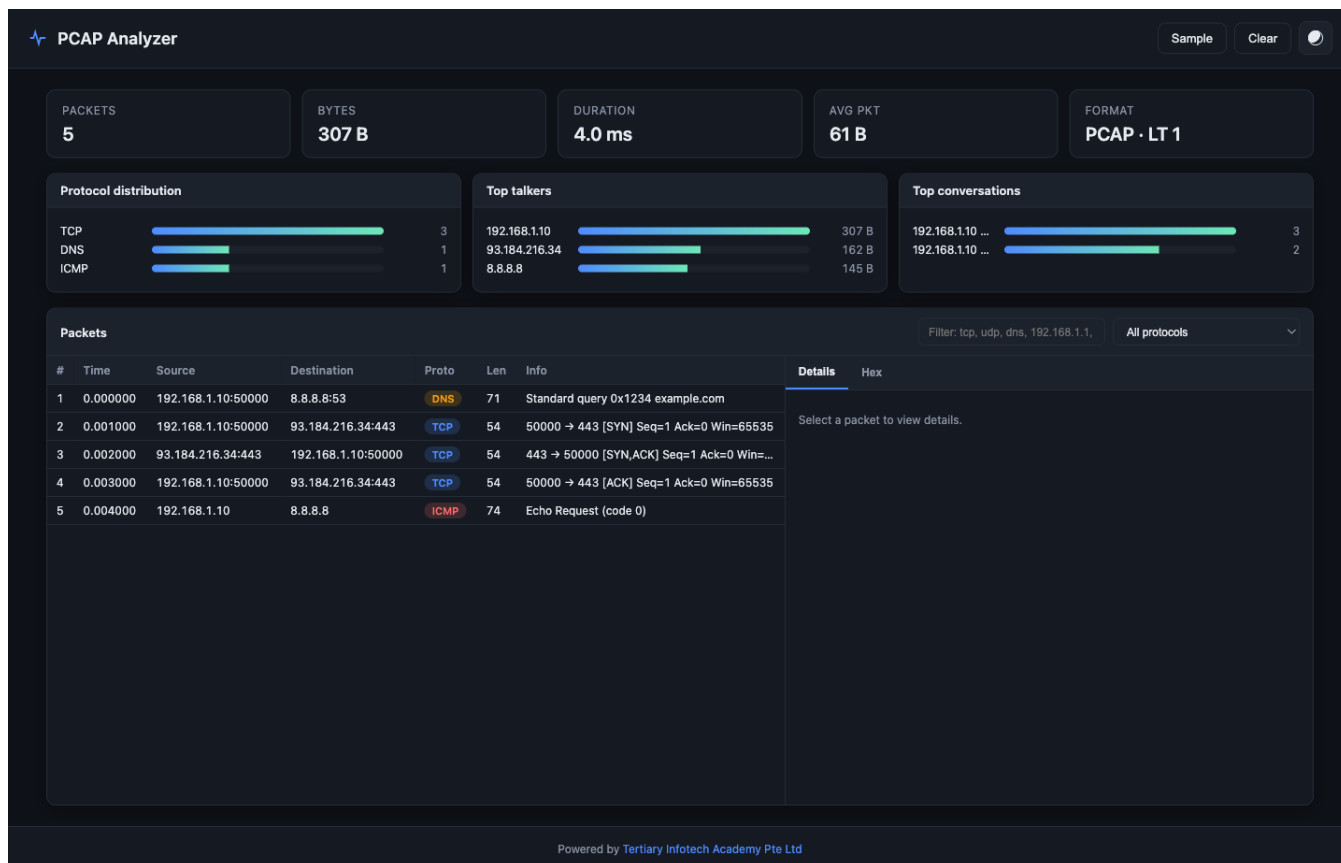
Netmask 11111111.11111111.11111111.00000000

Network 11000000.10101000.00000001.00000000

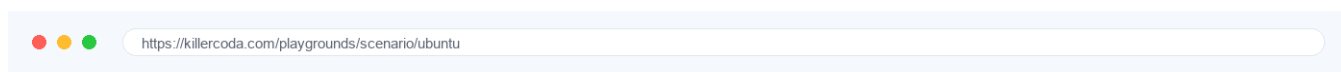
Broadcast 11000000.10101000.00000001.11111111

Wildcard 00000000.00000000.00000000.11111111

IP Calculator — the panels and fields this lab uses.



PCAP Analyzer — the panels and fields this lab uses.



Killercoda Ubuntu Playground — a real Linux machine in the browser

```
Terminal — root@controlplane

$ cat /etc/os-release | grep PRETTY
PRETTY_NAME="Ubuntu 22.04.3 LTS"
$ apt-get update -qq && apt-get install -y net-tools dnsutils
Reading package lists... Done
$ ip -brief addr show
lo        UNKNOWN  127.0.0.1/8
eth0      UP        10.244.0.14/24
$ dig +short www.tertiarycourses.com.sg
104.22.35.191
$ chmod 640 ~/applus/logs/system.log && ls -l ~/applus/logs/
-rw-r----- 1 root root 118 Aug 19 09:14 system.log
$ grep -c 'ERROR' ~/applus/logs/system.log
2
$ _
```

- 1 Open the URL and wait for the prompt. You get root on a real Ubuntu machine — nothing to install.
- 2 Start every lab with apt-get update, then install only the packages that lab needs.
- 3 The playground RESETS when idle. If your terminal disappears, reopen it and re-run the setup steps.

Killercoda Ubuntu Playground — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Build the diagnostic sequence bottom-up: physical link, IP addressing, default gateway, DNS resolution, then the application itself. Testing out of order produces misleading results.
2. Open the Killercodea Ubuntu playground and install the diagnostic tool set.

```
apt-get update -qq && apt-get install -y iproute2 dnsutils traceroute net-tools  
tcpdump curl
```

3. Layer 1 — confirm the interface is physically up and has carrier, since every layer above depends on it.

```
ip -brief link show
```

4. Layer 2 and 3 — record the interface addresses and identify whether the address is valid, or an APIPA 169.254 address meaning DHCP failed.

```
ip -brief addr show
```

5. Verify the address and mask in IP Calculator at <https://alfredang.github.io/ipcalculator/> and confirm the host sits inside its own subnet's usable range.
6. Layer 3 — confirm a default route exists, then test reachability to the gateway. No default route means no traffic can leave the subnet.

```
ip route show && ping -c 3 $(ip route | awk '/default/ {print $3}')
```

7. Test reachability beyond the gateway by IP address, which isolates routing from DNS entirely.

```
ping -c 3 8.8.8.8
```

8. Layer 7 — test DNS resolution separately, because a machine that pings an IP but not a name has a DNS fault and nothing else.

```
dig +short www.tertiarycourses.com.sg && cat /etc/resolv.conf
```

9. Trace the path to identify where latency is introduced or where the path stops.

```
traceroute -m 12 8.8.8.8
```

10. Check active connections and listening sockets to confirm the application layer is actually working.

```
ss -tunap | head -20
```

11. Capture traffic and analyse it in PCAP Analyzer when the commands are inconclusive, since the packets show what the tools only summarise.

```
tcpdump -i any -c 100 -w /root/netfault.pcap & sleep 3; curl -s https://example.com > /dev/null; wait; tcpdump -r /root/netfault.pcap -nn | head -20
```

12. Build the symptom map for APIPA address, no default gateway, pings IP but not name, high latency, jitter degrading VoIP, port flapping, intermittent wireless and limited connectivity — giving the layer, the test and the fix for each.

Test it — verification

Every command in the sequence runs successfully on the playground, the sequence tests layers strictly bottom-up, IP Calculator confirms the host address sits within its usable range, and the symptom map assigns a layer and a specific test to all eight faults.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the <code>apt-get install</code> step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against <code>labs/tools.md</code> . All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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